

Matthew Roberts

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Education:

Michigan State University – East Lansing, MI – B.S. in Data Science – May 2024 – 3.9 GPA

Summary:

Quantitative analyst with experience in commercial insurance pricing, underwriting analytics, and model-driven decision support. Known for working closely with stakeholders to define problems, translate analytics into clear recommendations, and drive decisions in complex, regulated environments.

Experience:

Predictive Modeler

May 2024 – Present | Auto-Owners Insurance – Lansing, MI

- Own commercial auto pricing and loss ratio improvements by partnering with actuarial and underwriting stakeholders to deliver frequency, severity, and combined GLMs (WTW Emblem) as part of a company-wide refresh.
- Build SQL and Excel analyses that integrate internal and third-party datasets (e.g., Polk, Experian, Equifax) to support risk segmentation, pricing decisions, and data-driven business recommendations.
- Support underwriting strategy by analyzing segment performance to recommend whether to pause underwriting in specific segments (e.g., ISO-rated, Lessors Risk Only) or pursue targeted premium growth.
- Present modeling results to actuarial teams and leadership, translating technical findings into clear implications for premium, underwriting strategy, and operational decisions.
- Serve as the primary analytics liaison between underwriters, system architects, and analysts in an initiative to improve data quality and policy rating accuracy by identifying missing values and developing models for imputation.
- Support internal AI pilots by designing labeling workflows and validating outputs to improve documentation/analyses.
- Reduce model validation turnaround from months to hours by automating visualization and model packet creation in Python, replacing slow manual workflows with a repeatable pipeline. Git based version control to support longevity.
- Review and support rate filings for commercial property models for dozens of states.

IT-Ambulatory Intern

May 2023 – August 2023 | Sparrow Hospital – Lansing, MI

- Improved caregiver report usability by cleaning and reorganizing large sets of clinical reports for clearer interpretation.
- Negotiated with nursing management and IT to determine report size and content.
- Supported clinical decision-making by coordinating to define report details like drug groupings and lab standards.
- Ensured secure and efficient workflows by quickly learning Epic and applying HIPAA guidelines when handling data.

Data-Analyst Intern

May 2022 – July 2022 | Cedar Point – Sandusky, OH

- Took initiative to help management monitor labor and identify scheduling issues by building Excel dashboards.
- Reduced manual workload by developing Python data pipelines to streamline dashboard updates for management.

Projects:

Pavilion Construction: Scoped, funded, and executed a 16'x16' pavilion build for Eagle Scout project. Aligned requirements with school, utility, and building code stakeholders, managed procurement and delivery schedule, and directed 20+ volunteers across 6–7 build days to deliver on schedule.

Municipality Scraper: Built an end-to-end ingestion pipeline (Python) to collect municipal documents (PDF/DOCX), store in B2, extract text (Google OCR/PDF parsing), and load into PostgreSQL for downstream analytics and feature extraction. Implemented LLM-assisted backfills to extract meeting dates and other structured fields from raw documents.